

CE 310 — Week 12 Pre-Lab Quiz

Week: 12 | Topic: Sensitivity Analysis Points: 10 questions × 1 pt each = 10 pts Submission: Complete in D2L before class begins. No partial credit for MC.

Instructions

Select the single best answer for each question. The quiz covers the definition and purpose of sensitivity analysis, one-at-a-time (OAT) perturbation, tornado diagrams, and how linear model coefficients relate to sensitivity. Review the Week 12 Background reading before class.

Questions

Q1. Sensitivity analysis in engineering models is primarily used to:

- (A) Determine whether the model is statistically significant
 - (B) Quantify how much the model output changes when one or more inputs change
 - (C) Replace regression with a simpler calculation method
 - (D) Eliminate uncertainty in the input data
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Q2. For a linear regression model $\hat{y} = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2$, the analytical sensitivity of $\hat{y}$ with respect to x_1 is:

- (A) β_0 (the intercept)
 - (B) β_1 (the coefficient on x_1), and it is constant for all values of x_1
 - (C) The ratio β_1/β_2
 - (D) It depends on the current value of x_1
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Q3. In a one-at-a-time (OAT) sensitivity analysis, you:

- (A) Change all inputs simultaneously and record the combined effect
 - (B) Change one input at a time while holding all others at their baseline values, and measure the output change
 - (C) Remove one predictor from the regression model and refit
 - (D) Compute the derivative of the output analytically using calculus
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Q4. The Model 2 regression result from Week 10 was: $\text{Cooling_kWh} = -56.87 + 1.026 \times \text{Outdoor_Temp_F} + 17.85 \times \text{BH}$

If outdoor temperature increases by 3°F (holding BH fixed), the predicted cooling changes by:

- (A) 1.026 kWh/hr (the raw slope, regardless of the 3°F)
 - (B) 3.078 kWh/hr (3×1.026)
 - (C) 17.85 kWh/hr (the BH coefficient)
 - (D) -56.87 kWh/hr (the intercept)
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Q5. A tornado diagram displays sensitivity results with:

- (A) A pie chart where each slice represents the fraction of variance explained by one predictor
 - (B) Horizontal bars sorted from the largest output swing (top) to smallest (bottom), one bar per input
 - (C) A line connecting minimum and maximum output values across all inputs simultaneously
 - (D) A histogram of prediction errors for each input scenario
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Q6. At the design point ($T = 85^\circ\text{F}$, $BH = 1$), a $\pm 10\%$ perturbation of each Model 2 parameter gives these swings: Outdoor_Temp_F coefficient = 17.44 kWh/hr, Intercept = 11.37 kWh/hr, BH coefficient = 3.57 kWh/hr. Which parameter should an engineer prioritize monitoring or refining, and why?

- (A) The BH coefficient, since it appears last in the model equation
 - (B) The Intercept, since it is the largest fixed value in the equation
 - (C) The Outdoor_Temp_F coefficient — it produces the largest swing in predicted output, so the model is most sensitive to it
 - (D) All three equally, since sensitivity analysis cannot rank inputs
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Q7. A $\pm 5^\circ\text{F}$ systematic bias in weather data affects 8,760 hourly predictions. Using the Model 2 slope of 1.026 kWh/ $^\circ\text{F}\cdot\text{hr}$, the total annual prediction error is approximately:

- (A) $5 \times 1.026 = 5.13$ kWh (only one hour matters)
 - (B) $8,760 \times 1.026 = 8,988$ kWh (ignoring the 5°F multiplier)
 - (C) $8,760 \times 5 \times 1.026 \approx 44,940$ kWh
 - (D) $1.026 / 8,760 \approx 0.0001$ kWh/hr
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Q8. The interaction Model 5 gives an occupied slope of 1.78 kWh/ $^\circ\text{F}$ and an unoccupied slope of 0.71 kWh/ $^\circ\text{F}$. Which statement correctly describes the sensitivity difference?

- (A) The occupied building's cooling prediction is $2.5\times$ more sensitive to outdoor temperature than the unoccupied building

- (B) The occupied building has a lower cooling prediction at every temperature than the unoccupied building
 - (C) The slopes are only different because of rounding; the true sensitivity is the same
 - (D) The unoccupied slope is more reliable for design purposes because it has a smaller magnitude
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Q9. An engineer uses the Model 2 slope of 1.026 kWh/°F to specify sensor accuracy. If a $\pm 1^\circ\text{F}$ sensor error should produce no more than a 2% error in the design-point prediction of 50 kWh/hr, can a standard temperature sensor (accuracy $\pm 0.5^\circ\text{F}$) meet this requirement?

- (A) No — a $\pm 0.5^\circ\text{F}$ sensor causes a $1.026 \times 0.5 = 0.513$ kWh/hr error, which exceeds 2% of 50 kWh/hr
 - (B) Yes — a $\pm 0.5^\circ\text{F}$ sensor causes a 0.513 kWh/hr error; $0.513/50 = 1.0\%$, which is within 2%
 - (C) The sensor accuracy cannot be determined from a regression model
 - (D) The question is irrelevant because temperature sensors do not affect energy predictions
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Q10. Sensitivity analysis and hypothesis testing (Week 11) answer different questions. Which pairing is correct?

- (A) Sensitivity analysis: “Is the BH coefficient statistically different from zero?” · Hypothesis testing: “How much does the prediction change when BH changes?”
 - (B) Sensitivity analysis: “How much does the prediction change when an input changes?” · Hypothesis testing: “Is an observed difference likely to have occurred by chance?”
 - (C) Both answer the same question — whether a predictor is important for the model
 - (D) Sensitivity analysis replaces hypothesis testing once the model R^2 exceeds 0.70
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